Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net force, velocity change, and final position in one model.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain the assumptions and limits of the prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “reference frame” and “net force” to explain your model or result. Sentence starters are optional.

reference frame: the origin and directions used to describe motion

net force: the vector sum of all forces on one object

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

A 3 kg box has μ_s=0.50 and μ_k=0.40 on a level floor; g=9.8.
Compare pushes of 6 N and 18 N, starting from rest.
Find actual friction and acceleration.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

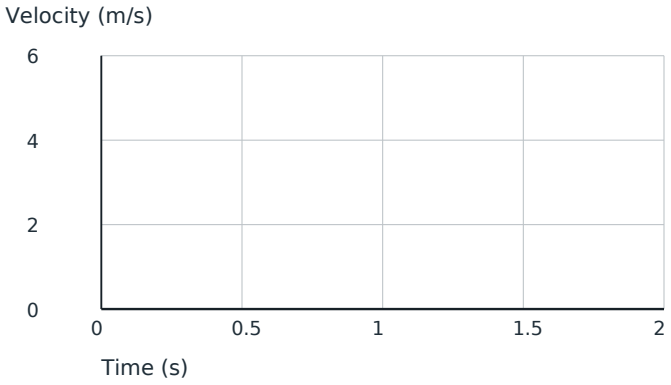
02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + ½(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A 2 kg cart starts from rest under 6 N right and 2 N left for 2 s. Net force 4 N gives $a=2$ m/s². Its displacement is $½\times2\times2^2=4$ m right.

Your turn: Draw the force diagram and the matching velocity-time triangle.
Check the 4 m using its area.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Build a complete prediction

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Continue down each column.

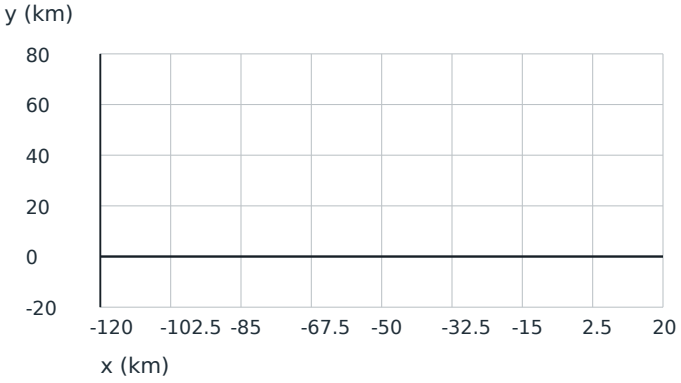
Sentence starters are optional. You may plan in any language.

03 End point versus displacement

In a simplified tracking model, a probe starts at (-100, 0) km and moves with constant velocity (4, 3) km/s for 20 s. The origin marks a stationary reference point.
Find final coordinates, displacement magnitude and final distance from the origin. Plot the route.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Add bounds

For the probe in question 3, the initial x coordinate has a bound of ±1 km and each velocity component has an independent bound of ±0.1 km/s. Initial y and time are exact.
Find the final x and y intervals.

Sentence starter: I found … [unit]. This value means … in this situation.

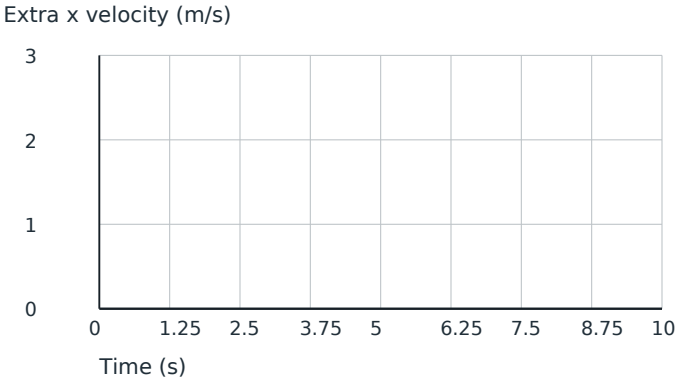
Coordinate	Central (km)	Lower (km)	Upper (km)
x	____	____	____
y	____	____	____

Marzano check - circle: ① ② ③

05 A new force

A 1000 kg probe experiences a constant 200 N net force in +x for 10 s. Compared with coasting from the same initial state, find the extra x velocity and extra x displacement.
Use an acceleration-time or velocity-change graph to check.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

Build a complete prediction

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Continue down each column.

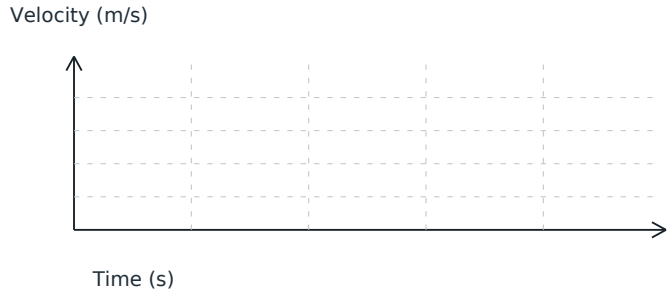
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A car starts at +35 m/s and brakes at -5 m/s² until rest. A student extends that acceleration to 9 s.
Find the stop time/distance and explain why extension is invalid for ordinary braking.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3

07 Exit check

Write a short public explanation of questions 3 and 4. Include the predicted position and coordinate ranges.
Explain two reasons this simplified model cannot establish a real asteroid collision forecast.

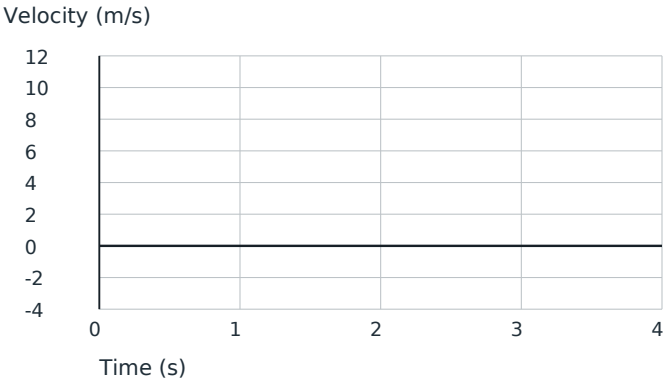
Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A cart starts at x=3 m with v=-2 m/s. Its mass is 2 kg and constant net force is +6 N for 4 s.
Predict final velocity and position.
Check whether it turns.

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: 1 2 3